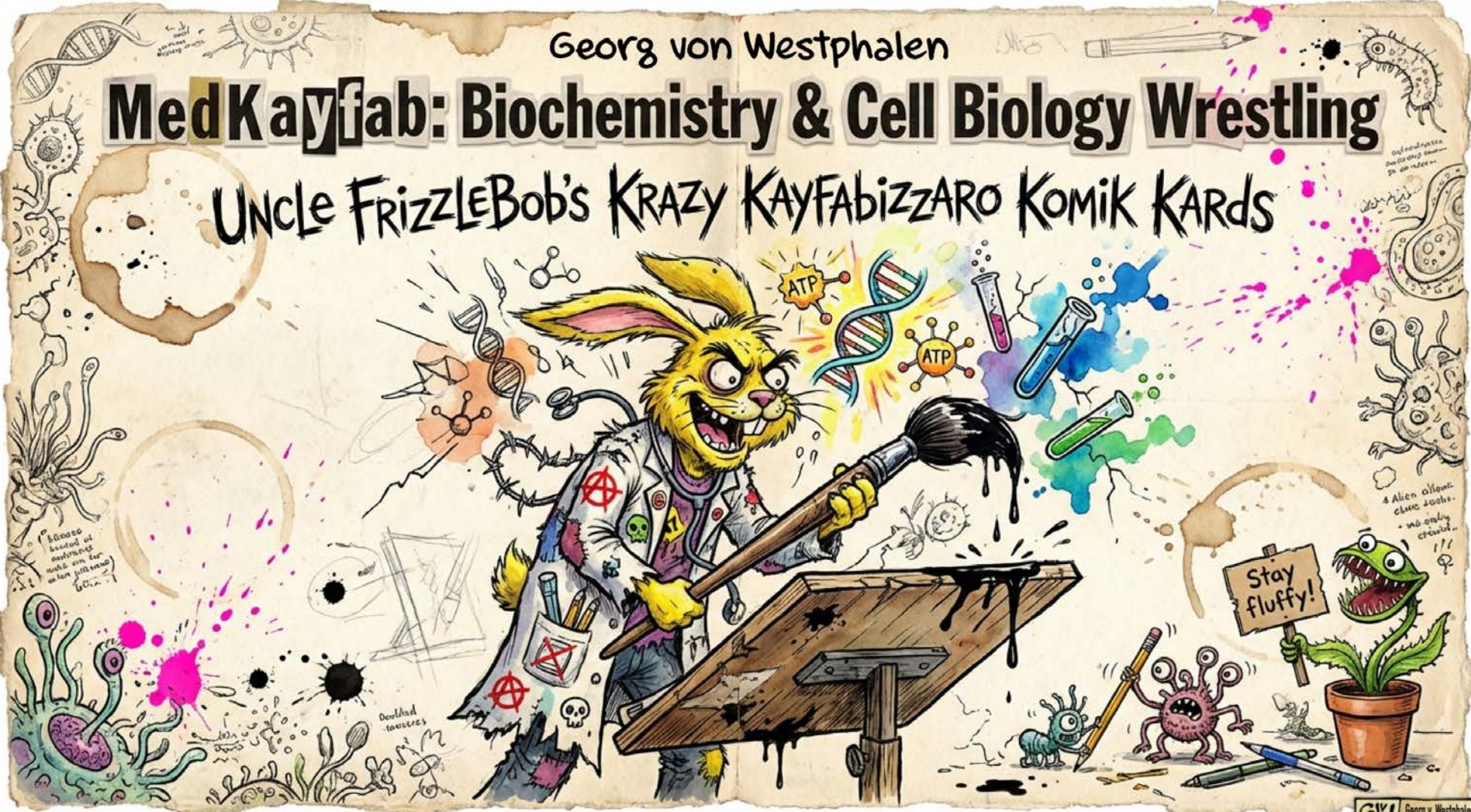


Georg von Westphalen

MedKayfab: Biochemistry & Cell Biology Wrestling

UNCLE FRIZZLEBOB'S KRAZY KAYFABIZZARO KOMIK KARDS



Membrana the Bilayer

Cell Membrane

STAT: 1 Kinked Tail = Fluidity



POWER: Flippase Swap — Burn ATP to move Phosphatidylserine (PS) to the inner leaflet.
LORE: Externalized PS is a macrophage eat me signal. Statins reduce cholesterol synthesis, altering membrane lipid composition and signaling.

Channel Charlie

Membrane Transport

STAT: 3 Na⁺ Out / 2 K⁺ In

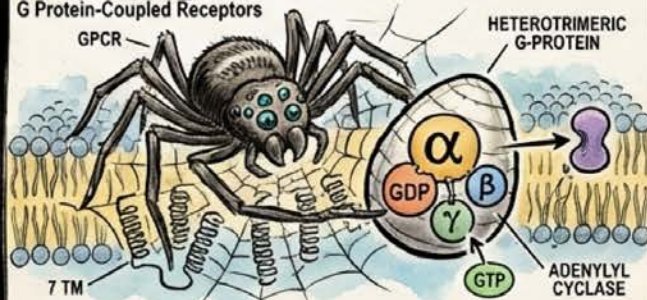


POWER: Primary Active Transport — Maintain the resting potential gradient to drive secondary co-transport.
LORE: Digitalis inhibits Na⁺/K⁺-ATPase → intracellular Na⁺ rises → Ca²⁺ extrusion via NCX drops → cardiac contractility increases.

GPCR Georgie

G Protein-Coupled Receptors

STAT: 7 Transmembrane Helices

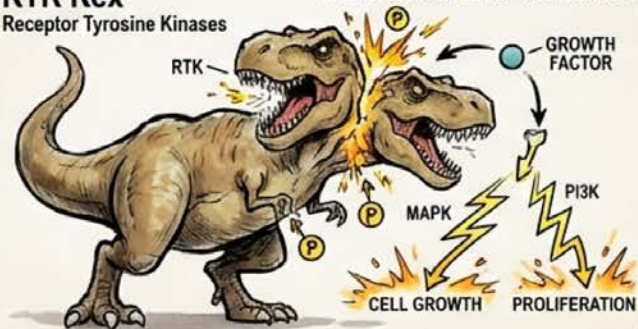


POWER: Second Messenger Swarm — Exchange GDP for GTP on Gα to activate adenylyl cyclase or PLC-β.
LORE: Cholera toxin ADP-ribosylates Gsα, locking it in the GTP-bound active state → massive cAMP → CFTR activation → fatal secretory diarrhoea.

RTK Rex

Receptor Tyrosine Kinases

STAT: 30% of Cancers = KRAS Mutant



POWER: Dimerize & Destroy — Bind extracellular growth factor to trans-autophosphorylate intracellular tyrosine residues.
LORE: Constitutive activation of the RAS/MAPK cascade via KRAS mutations renders colorectal tumors completely resistant to anti-EGFR therapies like cetuximab.

What the FLUFF?!

NF- κ B Nadia

Inflammation Master Regulator

STAT: p65/p50 Heterodimer



POWER: I κ B Degradation — Trigger the proteasome to destroy the I κ B inhibitor, freeing NF- κ B to enter the nucleus.
LORE: Glucocorticoids block this by upregulating I κ B α transcription and directly transrepressing p65. Boriszomb kills myeloma cells by blocking I κ B proteasomal degradation.

mTOR the Master Switch

STAT: Ser473 = Full Akt Activation
Nutrient Sensor



POWER: Anabolic Trigger — If amino acids and ATP are high, phosphorylate S6K1 to drive ribosome biogenesis.
LORE: Amino acids (especially leucine) recruit mTORC1 to the lysosomal surface. Rapamycin targets this complex to induce profound immunosuppression.

Wnt Wendy

Stem Cell Architect

STAT: 85% Sporadic CRC = APC Mutated



POWER: β -catenin Accumulation — Inhibit the GSK-3 β destruction complex to allow β -catenin to transcribe c-Myc and cyclin D1.
LORE: APC is the gatekeeper of colorectal cancer. Truncated APC fails to degrade β -catenin, driving the relentless adenoma-carcinoma sequence.

Hedgehog Harry

Developmental Signaler

STAT: 1 Primary Cilium Required



POWER: Smoothened Derepression — Bind Hedgehog ligand to internalize PTCH1 and free SMO to activate GLI transcription factors.
LORE: PTCH1 loss-of-function drives Basal Cell Carcinoma (BCC). Vismodegib shrinks these tumors dramatically by directly inhibiting the downstream SMO.

KayfaBIZARRO!

KayfaBIZARRO!

Notch Nina

Cell Fate Determination

STAT: S2 + S3 Cleavage



POWER: Lateral Inhibition — Upon adjacent ligand binding, release NICD to repress alternative cell fate genes.
LORE: NOTCH1 activating mutations drive >60% of T-ALL. γ -secretase inhibitors failed as Alzheimer's drugs due to extreme GI toxicity from goblet cell metaplasia.

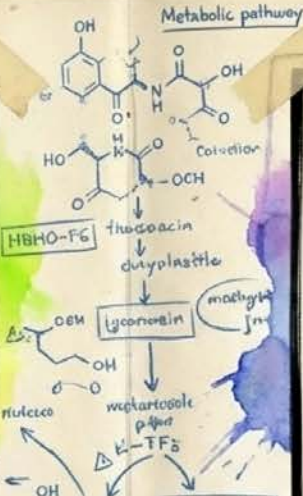
Glyco the Engine

STAT: 2 ATP + 2 NADH per Glucose



POWER: Warburg Shift — Override mitochondrial limits to preferentially run aerobic glycolysis for rapid ATP and biosynthesis.
LORE: PFK-1 is the rate-limiting step. ¹⁸F-FDG PET imaging exploits the Warburg effect by trapping phosphorylated glucose analogue (FDG-6-P) inside voracious tumors.

Metabolic pathway



Hypoxia HIF

STAT: 70–80% Clear Cell RCC = VHL Loss



POWER: VEGF Summon — Evade VHL-mediated ubiquitination to transcribe angiogenesis and glycolysis genes.
LORE: Vrl mutation stabilizes HIF-1 α regardless of oxygen, creating a highly vascular pseudo-hypoxic state in clear cell Renal Cell Carcinoma.

Krebs the Cycle

STAT: 3 NADH + 1 FADH₂ per Turn



POWER: NADH Generation — Oxidize Acetyl-CoA to CO_2 to capture high-energy electrons.
LORE: Thiamine (B1) deficiency halts α -KG dehydrogenase. Giving IV glucose without thiamine to an alcoholic precipitates fatal Wernicke's encephalopathy.

ETC Power Plant

Oxidative Phosphorylation

STAT: ~30-32 Net
ATP per Glucose



POWER: Proton Gradient — Pump H⁺ to the intermembrane space to create the electrochemical gradient for ATP synthase.

LORE: Cyanids halts Complex IV (cytochrome oxidase) by binding Fe²⁺. Treat with hydroxocobalamin and amyl nitrite to induce methaemoglobin as a cyanide decoy.

Beta-Ox Betty

Fatty Acid Oxidation

STAT: Palmitate =
~108 Net ATP



POWER: Carnitine Shuttle — Exchange acylcarnitine for free carnitine to breach the inner mitochondrial membrane.

LORE: MCAD deficiency prevents medium-chain fat oxidation. Fasting triggers lethal hypoketotic hypoglycaemia because alternative fuels cannot be synthesized.

PDH Gatekeeper

Pyruvate Dehydrogenase

STAT: 3 Enzymes
+ 2 Regulators



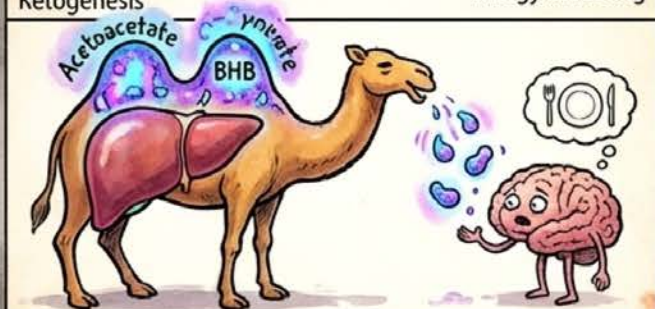
POWER: Irreversible Commitment — Convert pyruvate to Acetyl-CoA. Once crossed, carbons cannot return to glucose.

LORE: PDH deficiency causes severe lactic acidosis. A ketogenic diet bypasses PDH by supplying brain energy via β -hydroxybutyrate.

Ketone Kenny

Ketogenesis

STAT: 70% of Brain
Energy in Fasting



POWER: Brain Emergency Fuel — Convert excess Acetyl-CoA into soluble ketone bodies to bypass the blood-brain barrier.

LORE: The liver produces ketones but lacks SCOT to use them. In DKA, high NADH from alcohol oxidation increases BHB, masking ketones on a standard urine dipstick.

KayfaBONGO!



Glycogen Greta

Glycogenolysis

STAT: Branches every
8-12 Glucose



POWER: Phosphorylase Cleave — Remove terminal glucose as G-1-P in response to glucagon or epinephrine.

LORE: Von Gierke disease (G6Pase deficiency) traps glucose in the liver, causing fasting hypoglycaemia, lactic acidosis, and a doll-like facies.

Gluconeogenesis Gloria

Glucose Synthesis

STAT: 6 ATP Cost
per Glucose



POWER: Bypass Irreversibility — Use Pyruvate Carboxylase, PEPCK, FBPase-1, and G6Pase to force metabolism backward.

LORE: Metformin inhibits ETC Complex I, impairing the mitochondrial malate-aspartate shuttle and starving PEPCK of oxaloacetate, thus lowering fasting glucose.

FA Synth Fiona

Lipogenesis

STAT: 7 Cycles =
Palmitate (16:0)



POWER: Malonyl-CoA Addition — Dephosphorylate ACC via insulin to activate the rate-limiting step of lipogenesis.

LORE: Fatty acid synthesis strictly requires NADPH, linking it dependently to the pentose phosphate pathway and malic enzyme.

Cholesterol Carl

Mevalonate Pathway

STAT: 40-60% Extra LDL
Drop with PCSK9i



POWER: SREBP Feedback — If cholesterol falls, cleave SREBP to enter the nucleus and upregulate LDL receptors.

LORE: Statins competitively inhibit HMG-CoA reductase. This paradoxically lowers blood cholesterol primarily by triggering SREBP to upregulate hepatic LDL receptor clearance.

HUMBUG!

HUMBUG!

Urea Cycle Ulrich

Ammonia Disposal

STAT: 5 Steps
(2 Mito / 3 Cyto)



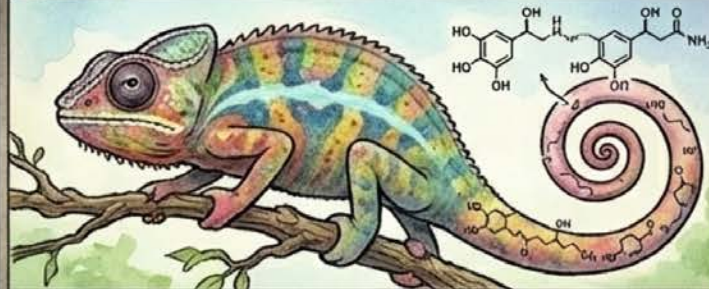
POWER: Nitrogen Excretion — Utilize CPS-I and NAG to commit toxic NH_3 to the urea cycle.

LORE: Hyperammonaemia drives glutamine accumulation in astrocytes, causing massive osmotic cerebral oedema. Treat CPS-I deficiency with arginine.

Phenylalanine Phil

Tyrosine Metabolism

STAT: 30% PKU =
BH4 Responsive



POWER: Tyrosine Conversion — Use PAH and BH4 to convert essential phenylalanine to tyrosine, melanin, thyroid hormone, and dopamine.

LORE: PKU (PAH deficiency) causes phenylalanine to cross the BBB, competitively inhibiting dopamine and serotonin synthesis, causing intellectual disability and hypopigmentation.

Porphyria Pete

Heme Synthesis

STAT: 50% PBG
Deaminase Activity



POWER: Heme Synthesis — Assemble protoporphyrin IX and load it with iron via ferrochelatase.

LORE: In Acute Intermittent Porphyria, CYP inducers (barbiturates, phenytoin) accelerate ALAS1, causing neurotoxic PBG buildup. Never give phenytoin for AIP seizures.

Purines & Pyrimidines Pablo

Nucleotide Salvage

STAT: Uric Acid
Crystals at pH < 5.8



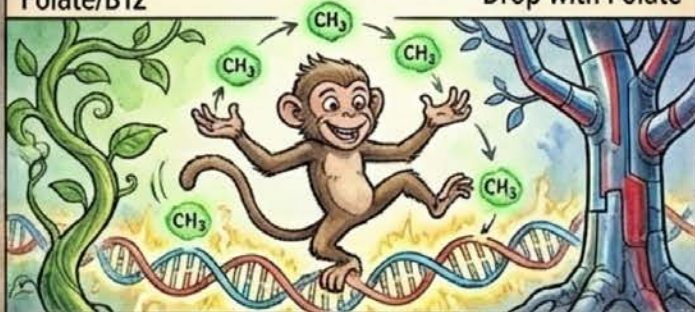
POWER: De Novo Salvage — Recycle hypoxanthine/guanine via HGPRT to save massive energy costs.

LORE: Lesch-Nyhan syndrome (absent HGPRT) forces purine degradation to uric acid, causing gout, and simultaneously destroys basal ganglia dopaminergic neurons causing self-mutilation.

One-Carbon Ollie

Folate/B12

STAT: 70% NTD
Drop with Folate



POWER: Methyl Trap — Trap THF as N5-methyl-THF if Vitamin B12 is absent.

LORE: B12 deficiency causes a functional folate deficiency (megaloblastic anaemia) PLUS methylmalonate accumulation that demyelinate the dorsal columns and corticospinal tracts.

Tryptophan Tyrone

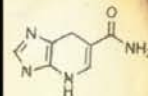
Dual Pathways

STAT: 60mg Tryptophan
= 1mg Niacin



POWER: Serotonin/NAD⁺ Shift — Divert 95% of substrate via the kynurenine pathway to generate immune tolerance and NAD⁺.

LORE: Midgut carcinoid tumors rarely cause systemic carcinoid syndrome until they metastasize to the liver, bypassing hepatic MAO clearance.



KayfaBIZARRO!



G6PD Godfrey

Oxidative Stress

STAT: Class II Favism
= <10% Activity



POWER: NADPH Regeneration — Run the oxidative pentose phosphate pathway to keep glutathione in a reduced, protective state.

LORE: G6PD deficiency depletes NADPH. Oxidative stress (primaquine, fava beans) oxidizes hemoglobin into Heinz bodies, which splenic macrophages bite out, causing haemolytic anaemia.

Lysosome Lenny

Storage Diseases

STAT: Optimal
pH 4.5–5.0



POWER: Substrate Accumulation — Fail to hydrolyze lipids, causing intralysosomal inclusions that progressively kill the cell.

LORE: Tay-Sachs (HEXA deficiency) causes GM2 ganglioside accumulation in CNS neurons, presenting as a cherry-red spot at the macula and fatal neurodegeneration.



DNA Rep Dina

Replication

STAT: 1 Error per
10 Billion bp



POWER: Semi-Conservative Replication — Use DNA Pol δ and ϵ to ensure high-fidelity genome duplication.

LORE: Fluoroquinolones (ciprofloxacin) are bactericidal because they inhibit bacterial topoisomerase II (DNA gyrase), preventing the relaxation of supercoiling ahead of the replication fork.

DNA Repair Rex

Repair Pathways

STAT: 10,000 BER
Lesions/Cell/Day



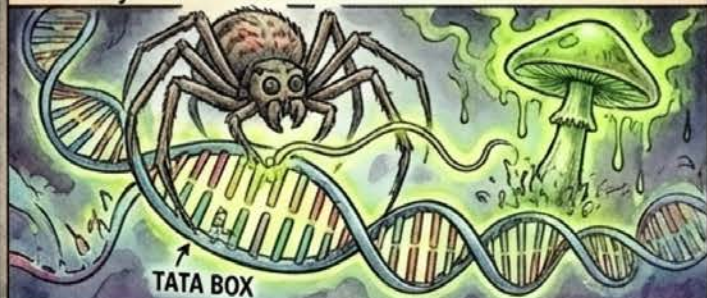
POWER: Synthetic Lethality — Trap PARP onto DNA to force a double-strand break that BRCA-mutant cells cannot repair.

LORE: Lynch syndrome is caused by germline mismatch repair (MMR) gene mutations, causing microsatellite instability (MSI) and massive colorectal/endometrial cancer risk.

Transcription Tina

mRNA Synthesis

STAT: 52 Heptapeptide
CTD Repeats



POWER: Promoter Binding — Recruit RNAPII to CpG islands to transcribe DNA into pre-mRNA.

LORE: Alpha-amanitin from the death cap mushroom specifically blocks the elongating Ser2-P form of RNAPII, causing irreversible, delayed hepatocyte death.

Translation Tommy

Ribosomes

STAT: 70S (Bacteria)
vs 80S (Human)



POWER: Peptide Bond Formation — Use the peptidyltransferase ribozyme center to link amino acids.

LORE: Aminoglycosides bind the 30S subunit causing misreading and bactericidal death, whereas macrolides bind the 50S exit tunnel causing bacteriostatic arrest. Diphtheria toxin halts eukaryotic eEF2.



Cell Cycle Celia

STAT: 3 Major Surveillance Points



POWER: Rb Phosphorylation — Hyperphosphorylate pRb via CDK4/6-cyclin D to release E2F and irreversibly commit to S phase.
LORE: CDK4/6 inhibitors (palbociclib) force a G1 arrest in ER+ breast cancer by preventing pRb phosphorylation, restoring the gatekeeper function.

Mitosis Mia

STAT: 1 Kinetochore per Chromatid



POWER: Chromosome Segregation — Satisfy the Spindle Assembly Checkpoint to cleave cohesin and trigger anaphase.
LORE: Taxanes (paclitaxel) stabilize microtubules preventing depolymerization, while vinca alkaloids prevent polymerization. Both abnormally alter tension, permanently arresting the cell in M-phase.

p53 Guardian

Tumor Suppressor

STAT: >50% Cancers Mutated



POWER: Apoptotic Arrest — If DNA damage is unreparable, transcribe BAX/PUMA for intrinsic apoptosis.
LORE: MDM2 is an E3 ligase that constantly degrades p53. Hotspot mutations in the p53 DNA-binding domain exert dominant-negative oncogenic effects.

Autophagy Ada

Self-Digestion

STAT: 1 Double-Membrane Phagophore

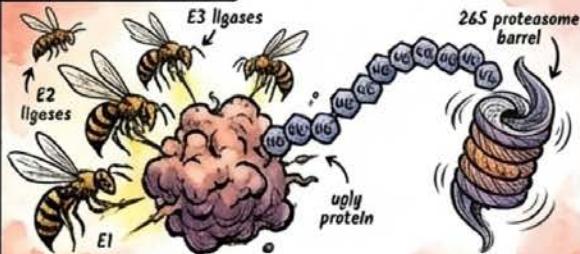


POWER: Phagophore Nucleation — Inhibit mTORC1 to activate ULK1 and lipidation of LC3-II to autophagosome membranes.
LORE: PINK1 and Parkin mediate mitophagy of depolarized mitochondria. Mutations in these fall to clear toxic mitochondria, causing autosomal recessive Parkinson's disease.

Ubiquitin Uri

Proteasome System

STAT: 76
Amino Acid Tag



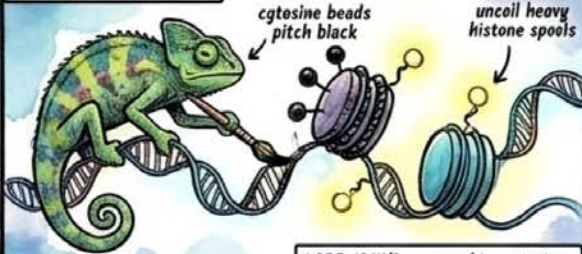
POWER: Proteasomal Targeting — Attoch E98-linked peigubigoitia chains to mark substrates for the 26S proteasome barrel.

LORE: PROTAC drugs are bifunctional molecules that artificially drag undruggable target proteins to an E3 ligase, hijacking Uri to destroy them.

Epigenetics Eddie

DNA/Histone Mod.

STAT: 15-20%
AML = DNMT3A Mutant



POWER: Chromatin Remodeling — Acetglate bistenes to open chromatin; methglate CpG islands to permanently silence tumor suppressors.

LORE: IDH1/2 neomorphic mutations produce 2-hydroxylglutarate (an oncometabolite) that competitively inhibits TET, causing DNA hypermethylation and a leukaemic differentiation block.

MicroRNA Mira

Non-Coding RNA

STAT: 19-22
Nucleotides



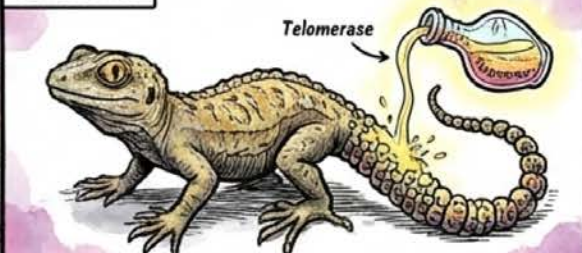
POWER: Translational Repression — Load the RISC complex to imperfectly kind target mRNAs, halting translation or triggering decay.

LORE: The ceRNA hypothesis states that long non-coding RNAs act as sponges, soaking up miRNAs to de-repress target oncogenes. miR-21 is a ubiquitous oncomiR.

Telomere Talia

Senescence

STAT: ~50 Divisions
(Hayflick Limit)



POWER: TERT Reactivation — Express reverse transcriptase to extend the 3' overhang, evading permanent replicative senescence.

LORE: TERT promoter mutations (creating de novo ETS binding sites) are the most common non-coding mutations in cancer (melanoma, GBM), granting replicative immortality.

ER Stress Ernie

Unfolded Protein Response

STAT:
14-Sugar Oligosaccharide Tag



POWER: Unfolded Protein Response — Halt global translation via PERK/eIF2 α while expanding ER chaperone capacity to restore proteostasis.

LORE: Multiple Myeloma cells survive at the brink of ER stress due to massive IgG secretion. Bortezomib pushes them over the edge into terminal UPR apoptosis.

Golgi Gloria

Vesicle Trafficking

STAT:
3 Cisternae (Cis, Medial, Trans)



POWER: M6P Tagging — Add mannose-6-phosphate in the cis-Golgi to route acid hydrolases exclusively to lysosomes.

LORE: In I-cell disease (GlcNAc-1-phosphotransferase deficiency), the M6P tag is never added. Hydrolases are secreted into blood instead, causing multi-system lysosomal storage failure.

Mitochondria Max

Mitochondrial Genetics

STAT:
16,569 bp Circular mtDNA



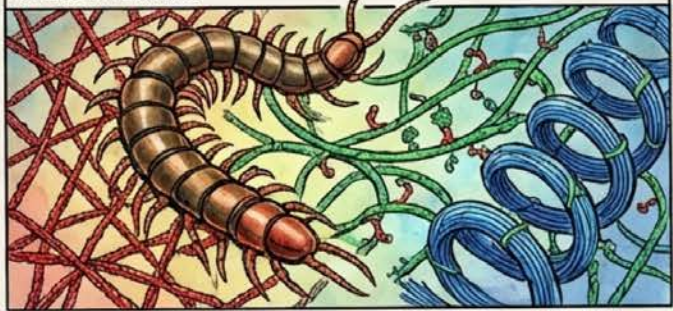
POWER: Heteroplasmic Shift — Segregate mutant mtDNA randomly during division; clinical severity depends entirely on the mutant-to-wild-type ratio threshold.

LORE: MELAS (m.3243A>G) impairs mitochondrial tRNAs, starving the brain of ATP and causing stroke-like episodes in non-vascular distributions, strictly via maternal inheritance.

Cytoskeleton Clio

Structural Filaments

STAT:
9+2 Microtubule Axoneme



POWER: Dynamic Instability — Rapidly alternate between GTP-driven polymerization and GDP-driven catastrophe to search intracellular space.

LORE: Mutations in tissue-specific intermediate filaments cause distinct diseases: Keratin (epidermolysis bullosa), Desmin (cardiomyopathy), and Lamin A (Progeria/rapid aging).

A octadecanoyl peptidyl transferase mutation can substantially disrupt protein folding pathways due to wrong cellular transport rates.



KayfaBINGO!

GW

Georg v. Westphalen

Peroxisome Perry

STAT: >C22 (VLCFA Threshold)

VLCFA Oxidation



POWER: Alpha/Beta Oxidation — Import and break down very-long-chain and branched fatty acids that mitochondria cannot process.

LORE: X-linked Adrenoleukodystrophy (ABCD1 mutation) traps VLCFAs outside the peroxisome, causing toxic accumulation that destroys the adrenal cortex and cerebral myelin.

Cilia Felix

STAT: 10–15 Hz Tracheal Beat

Motile & Primary Cilia



POWER: Mechanosensory Swoop — Move extracellular fluid or sense tubular flow via Polycystin 1/2 complexes to prevent aberrant proliferation.

LORE: Primary Ciliary Dyskinesia (dynein arm mutation) halts embryonic leftward nodal flow, meaning situs inversus occurs randomly in exactly 50% of patients.

Storage Sammy

STAT: 0% Aldolase B in HFI

Carb Metabolism Errors



POWER: Substrate Toxicity — Trap phosphorylated sugars intracellularly, depleting ATP and causing severe hepatotoxicity.

LORE: Classic galactosemia (GALT deficiency) causes neonatal liver failure, cataracts, and a highly specific vulnerability to fatal E. coli sepsis due to impaired neutrophil killing.

Wilson's Wally

STAT: >100µg/24h Urine Copper

Copper Transport



POWER: Biliary Failure — Mutate ATP7B so copper cannot lead into ceruloplasmin or excrete into bile, forcing toxic systemic spillover.

LORE: Serum ceruloplasmin is LOW in Wilson's disease because without ATP7B to load copper, unstable apoceruloplasmin is secreted and rapidly degraded in blood.

What the FLUFF?!



Acidaemias Otto

BCAA Catabolism

STAT: ~10% BCKDH Restored by Transplant



POWER: Keto-Acid Accumulation — Block BCKDH to force neurotoxic branched-chain keto acids into the blood and brain.

LORE: Propionic and Methylmalonic acidaemias cause profound hyperammonaemia because toxic propionyl-CoA directly inhibits CPS-I in the urea cycle.

CDG

Congenital Disorders of Glycosylation

STAT: 2-Peak IEF = CDG-Ia



POWER: Glycan Misfire — Fail to assemble N-linked oligosaccharides in the ER, devastating protein folding, secretion, and receptor function system-wide.

LORE: PMM2-CDG (Type Ia) presents with ataxia, coagulopathy, and inverted nipples. Underglycosylation of transferrin (type I pattern on IEF) is the delectable screening diagnostic.

Cancer Metabolism Marcus

Anapleresis

STAT: 18F-FDG = Warburg Tracer



POWER: Anaplerotic Theft — Siphon TCA intermediates for rapid biosynthesis, strictly requiring glutaminolysis to replenish the cycle.

LORE: Cancer cells are not just anaerobic; MPC and HIF-1 α rewire metabolism to actively use aerobic glycolysis and glutamine for massive biomass generation.

ROS Rosa

Oxidative Stress & Ferroptosis

STAT: 1-5% ETC Electron Leak



POWER: Lipid Peroxidation — Deplete glutathione (GSH) to disable GPx4, unleashing iron-dependent ROS to destroy the plasma membrane.

LORE: Ferroptosis is a distinct, non-apoptotic cell death driven by lipid peroxides. It is triggered when GSH is depleted and GPx4 falls, requiring Fenton-reaction iron.

BLÖDSINN!

Proteostasis Patricia

Protein Aggregation

STAT: 265
Proteasome Core



POWER: Prion-like Spread — Escape chaperone surveillance to form toxic aggregates (Lewy bodies, plaques) that spread across post-mitotic networks.

LORE: Parkinson's disease spreads via the prion-like misfolding of α -synuclein into Lewy bodies. CHIP normally links chaperones to the proteasome to prevent this.

KayfaBIZARRO!

KayfaBIZARRO!

Metabolic Syndrome Marco

Insulin Resistance

STAT: Waist ≥ 94 cm
(M) / ≥ 80 cm (F)



POWER: Ectopic Lipid Deposition — Overload visceral fat capacity to force toxic free fatty acids into liver and muscle, disabling insulin receptors.

LORE: Visceral adiposity triggers FFA release and ceramide accumulation, causing IRS-1 serine phosphorylation that fundamentally breaks the PI3K/Akt insulin signaling pathway.

The Biochem Attending

Integration

STAT: 100% Clinical
Correlation Required



POWER: Integrated Clinical Reasoning — Connect the molecular defect directly to the patient's presentation.

LORE: No shortcut to pattern recognition is permitted — explain the mechanism.

Case 1: Fasting hypoglycaemia = MCAD. Case 2: OCP + Rifampicin = AIP.

Case 3: Myeloma + Bortezomib = Terminal UPR ER Stress.

FrizzleBob's Golden Ticket

The Joker

STAT: 0 Filler



POWER: Narrative Memory — You remember stories, not lists.

LORE: The facts are real. The biology is flawless. The delivery is anarchy.
Now go sabotage the boards. HUMBUG!